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(Autonomous Institution, Affiliated to VTU, Belagavi)



A Project Report On

**EXPLAINABLE AI FOR DIABETIC RETINOPATHY
SCREENING IN RURAL AREA**

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CERTIFICATE

This is to certify that the project work entitled "**Explainable AI for Diabetic Retinopathy Screening in Rural India**" carried out by

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(Team **CODEX**), Bonafide students of the Department of Information Science and Engineering, Alva's Institute of Engineering and Technology, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Engineering in DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING of the VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI during the year 2025– 2026.** is a record of original work done under my guidance. It is certified that all corrections and suggestions indicated during internal assessment have been incorporated in the report. The project report satisfies the academic requirements in respect of project work prescribed for the said degree.

DECLARATION

We, the undersigned students of the Department of Information Science, ALVAS INSTITUTE OF ENGINEERING AND TECHNOLOGY, hereby declare that the project work entitled "**Explainable AI for Diabetic Retinopathy Screening in Rural India**" is an original work carried out by us under the guidance of Anjith Raja R - Trainer , Department of Information Science and Engineering.

We further declare that this work has not been submitted, in part or in full, to any other university or institution for the award of any degree or diploma. Wherever the work of others (publications, datasets, software libraries) has been used, it has been duly acknowledged and cited in the References.

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ABSTRACT

Diabetic retinopathy (DR) is a microvascular complication of diabetes and a leading cause of preventable vision loss among working-age adults. India has one of the largest diabetic populations in the world, while the availability of ophthalmologists in rural regions is extremely limited, which makes routine manual screening of every diabetic patient impractical. Artificial intelligence (AI) can support large-scale screening, but many existing systems behave as black boxes, are rarely designed for the variable image quality produced by portable fundus cameras in field conditions, and offer clinicians little insight into why a particular grade was predicted.

This project proposes **an explainable, quality-aware, MATLAB-based pipeline for AI-assisted diabetic retinopathy screening** intended for primary healthcare settings in rural India. The pipeline accepts a retinal fundus image and first performs image quality assessment (focus, illumination and field of view), rejecting ungradeable images with recapture feedback and applying adaptive enhancement to borderline images. The image is then preprocessed using resizing, denoising, illumination normalisation and Contrast Limited Adaptive Histogram Equalisation (CLAHE). Retinal structures and lesions, namely the optic disc, blood vessels, microaneurysms, exudates, haemorrhages and neovascularisation, are analysed to provide lesion-level evidence. A deep convolutional neural network, trained on public datasets such as APTOS 2019 and IDRiD, grades the image on the five-level international clinical DR severity scale (No DR to Proliferative DR) and produces a calibrated confidence score. Explainability is provided through Grad-CAM heatmaps and lesion overlays that are compiled into an annotated screening report, so that a healthcare worker or ophthalmologist can review the AI's reasoning quickly. A Simulink model of the telemedicine workflow is proposed to study acquisition rate, bandwidth, processing throughput and specialist review capacity for district-level programmes.

The design targets, taken from the project brief, are a sensitivity above 90% and specificity above 85% for referable DR (moderate or worse). These are **targets, not achieved results**; actual performance is reported as [XX%] until experiments are completed. The system is intended strictly as an AI-assisted screening and triage aid to support, and not to replace, qualified medical professionals, and it requires clinical validation before any real-world deployment.

Keywords: Diabetic Retinopathy, Explainable AI, Grad-CAM, Fundus Image, Deep Learning, Image Quality Assessment, MATLAB, Simulink, Rural Healthcare.

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ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the (ROC) Curve
CAM	Class Activation Mapping
CLAHE	Contrast Limited Adaptive Histogram Equalisation
CNN	Convolutional Neural Network
CPU / GPU	Central / Graphics Processing Unit
DFD	Data Flow Diagram
DL	Deep Learning
DME	Diabetic Macular Edema
DR	Diabetic Retinopathy
ECE	Expected Calibration Error
FN / FP	False Negative / False Positive
FOV	Field of View
Grad-CAM	Gradient-weighted Class Activation Mapping
HMA / MA	Haemorrhages and Microaneurysms / Microaneurysm
IDRiD	Indian Diabetic Retinopathy Image Dataset
ML	Machine Learning
NPDR / PDR	Non-Proliferative / Proliferative Diabetic Retinopathy
OD	Optic Disc
PHC	Primary Health Centre
RGB	Red, Green, Blue (colour model)
ROC	Receiver Operating Characteristic
ROI	Region of Interest
TN / TP	True Negative / True Positive
UI	User Interface
USN	University Seat Number
XAI	Explainable Artificial Intelligence

CHAPTER 1 – INTRODUCTION

1.1 Background

Diabetes mellitus is a chronic metabolic disorder whose long-term complications affect the eyes, kidneys, nerves and cardiovascular system. According to the project brief, India has over 77 million adults with diabetes, the second highest number in the world (**verify against the current IDF Diabetes Atlas**). As the diabetic population grows and ages, the number of people at risk of eye complications grows with it. Healthcare in India, however, is unevenly distributed: specialist services are concentrated in cities, whereas a large share of the population lives in rural areas served by primary health centres (PHCs) with limited diagnostic capability.

Medical imaging combined with AI has emerged as a practical approach to extend specialist-level screening to such settings. A fundus camera, including low-cost portable devices, captures an image of the retina that a trained observer, or an algorithm, can grade for signs of disease. This project develops such an approach with a distinguishing emphasis on **explainability** and **robustness to image quality**, two properties that are essential for trust and usability in the field.

1.2 Diabetic Retinopathy

Diabetic retinopathy is damage to the small blood vessels of the retina caused by prolonged high blood glucose. In the early stage the disease is usually **asymptomatic**, so patients often present only after vision has been affected. The disease progresses through recognisable stages:

- **Non-proliferative DR (NPDR):** characterised by microaneurysms (small red dots), intraretinal haemorrhages, hard exudates (lipid deposits), cotton-wool spots and venous beading or intraretinal microvascular abnormalities in the more severe stages.
- **Proliferative DR (PDR):** characterised by neovascularisation (growth of abnormal new vessels) on the optic disc or elsewhere, which can lead to vitreous haemorrhage and tractional retinal detachment.
- **Diabetic macular edema (DME):** swelling of the macula that can occur at any DR stage and threatens central vision. (DME grading is outside the primary scope of this project but is noted as future work.)

The **International Clinical Diabetic Retinopathy Disease Severity Scale** [1] describes five levels: no apparent retinopathy, mild NPDR, moderate NPDR, severe NPDR and proliferative DR. These map to Levels 0–4 used by the datasets discussed in this report. In screening programmes, "**referable DR**" generally means moderate NPDR or worse (Level 2 and above), and this definition is used throughout this report, consistent with the project brief.

According to the project brief, DR affects about 18% of India's diabetic population and timely screening and treatment can prevent a large majority of vision loss (**figures as stated in the brief; verify with cited sources before final submission**).

1.3 Problem Context in Rural India

1. **Specialist shortage.** The project brief states that there is roughly one ophthalmologist per 100,000 rural population (**verify**). Manual grading of every diabetic patient by a specialist is therefore infeasible.

2. **Distance and cost.** Patients must travel to district or city hospitals; loss of wages, travel cost and lack of symptoms lead to missed follow-ups.
3. **Variable image quality.** Images acquired by field staff using portable or handheld cameras suffer from defocus, uneven illumination, small pupils, media opacity and partially visible retinas. Systems trained only on clean hospital images often fail on such images.
4. **Limited connectivity.** Bandwidth in rural areas is uncertain, which affects telemedicine workflows.
5. **Trust and adoption.** Healthcare workers and ophthalmologists are reluctant to act on a bare label such as "Grade 3" without understanding the reasoning.

1.4 Role of Artificial Intelligence in DR Screening

Deep learning, particularly convolutional neural networks (CNNs), has achieved expert-level performance for DR detection in several published studies [2], [3], [4]. AI can (i) pre-screen large numbers of images automatically, (ii) prioritise patients who need urgent referral, and (iii) reduce the workload on specialists by removing clearly normal cases from the review queue. Field studies in India and other countries have also shown that real-world deployment raises practical problems such as image gradeability and workflow integration [5], [6], [7]. AI in this project is therefore positioned as a **screening and triage aid**, not a diagnostic authority.

1.5 Need for Explainable AI

A deep network normally outputs only a class and a probability. In medicine this is insufficient because:

- clinicians are accountable for decisions and need to verify the evidence;
- a model may rely on artefacts (camera edges, illumination patterns, dataset-specific cues) instead of true lesions, and such shortcuts are invisible without explanation;
- regulators and hospitals increasingly expect transparency for AI-assisted decisions;
- understandable explanations help healthcare workers explain the referral advice to patients.

Explainable AI (XAI) techniques such as Grad-CAM [8] highlight the image regions that most influenced a prediction. The literature also cautions that such visual explanations must themselves be tested and are not proof of clinical correctness [9], [10]; this report addresses that caution in Chapter 8 and Chapter 14.

1.6 Motivation

- The scale of the diabetic population versus the number of available specialists.
- The preventable nature of most DR-related blindness when detected early.
- The gap between research-grade AI and systems that are robust, transparent and usable at primary-care level.
- The opportunity to combine classical image-processing knowledge (quality checks, lesion detection) with deep learning and XAI in a single teachable, MATLAB-based pipeline.

1.7 Problem Statement

To design and prototype an explainable, quality-aware retinal image analysis pipeline that assists healthcare workers in screening for diabetic retinopathy in rural India by assessing fundus image quality, analysing retinal structures and lesions, grading DR severity on the international five-level scale with a calibrated confidence score, and presenting visual and lesion-level explanations in an automated screening report to support human-in-the-loop clinical review.

1.8 Objectives

1. To study public fundus image datasets (APTOS 2019, IDRiD) and prepare them for development and evaluation.
2. To develop an image quality assessment module that evaluates focus, illumination and field of view, and gives accept / enhance / recapture feedback.
3. To develop preprocessing and enhancement (denoising, illumination normalisation, CLAHE) suited to fundus images.
4. To implement analysis of the optic disc, blood vessels, microaneurysms, exudates, haemorrhages and neovascularisation to provide lesion-level evidence.
5. To train and evaluate a deep learning model for five-level DR severity grading, with the design targets of **sensitivity > 90% and specificity > 85% for referable DR** (targets from the project brief).
6. To provide Grad-CAM heatmaps, lesion overlays and calibrated confidence scores, and to generate an annotated screening report designed for review within about 30 seconds.
7. To model the telemedicine screening workflow in Simulink to study resource allocation for district-level programmes.
8. To document limitations honestly and state the need for clinical validation.

1.9 Scope of the Project

In scope: retrospective development and evaluation on public datasets; MATLAB-based prototype; image quality assessment; preprocessing; lesion/structure analysis; five-level grading; Grad-CAM and lesion-based explanation; report generation; Simulink workflow simulation; proposed user interface.

Out of scope: definitive diagnosis, treatment recommendation, prospective clinical trials, regulatory approval, DME grading (future work), and production-grade deployment. The prototype has **not** been clinically validated.

CHAPTER 2 – LITERATURE SURVEY

2.1 Overview

This chapter reviews research areas relevant to the project. Only publications that the team believes to be genuine are listed; **each entry must be re-verified (authors, year, venue, page numbers, figures) against the original paper before submission**. Quantitative claims are deliberately not reproduced here to avoid misquotation; the team should add them from the original papers.

2.2 AI-Based Diabetic Retinopathy Detection

Early automated DR systems used hand-crafted features (lesion counts, vessel measures) and classical classifiers such as SVMs. From 2016 onward, deep CNNs trained on large labelled fundus collections showed performance comparable to graders in retrospective validation [2], [3], [4]. Prospective and clinic-based studies, including an autonomous system evaluated in US primary care [11], an evaluation in Indian settings [5], and smartphone-based fundus photography with AI in India [7], indicate the promise as well as the practical challenges.

2.3 CNN-Based Retinal Image Classification

Modern backbones such as ResNet [12], DenseNet [13] and EfficientNet [14], normally pre-trained on natural images and fine-tuned (transfer learning), are widely used for DR grading, particularly on the APTOS 2019 competition data [15]. Ordinal nature of the labels ($0 < 1 < 2 < 3 < 4$) is often exploited through regression or ordinal losses.

2.4 Fundus Image Preprocessing

Fundus images show non-uniform illumination, low contrast in the green channel and camera noise. Commonly used techniques include cropping the black border, resizing, green-channel extraction, background subtraction/illumination correction, denoising and contrast enhancement using CLAHE [16], [17].

2.5 DR Lesion Detection

Microaneurysms, haemorrhages, hard exudates and soft exudates are detected by classical morphology-based and, more recently, deep segmentation methods such as U-Net [18]. The IDRiD challenge provides pixel-level lesion annotations that enable evaluation of these tasks [19], [20].

2.6 Explainable AI in Medical Imaging

Saliency methods [21], Grad-CAM [8] and related techniques visualise model evidence. Studies have warned that saliency maps may be insensitive to the model's parameters and can mislead [22], that interpretable models should be preferred for high-stakes decisions where possible [9], and that current XAI in healthcare offers no guarantee of correctness [10].

2.7 AI-Assisted Screening in Resource-Limited Settings

A human-centred evaluation of a deep learning DR system in clinics in Thailand documented that poor image quality, network delays and workflow mismatch reduced usefulness in practice [6]. Work on smartphone-based fundus photography with AI in India [7] and on comparing AI with manual grading in India [5] is directly relevant to the target setting of this project. Reliable probability estimates are also important; modern networks tend to be over-confident and can be recalibrated [23].

2.8 Summary Table

Table 2.1: Summary of Literature Reviewed (*Method, dataset and limitations are stated at a general level; add exact figures from the original papers [TO BE UPDATED]*)

Paper / Study	Year	Method	Dataset	Key Findings	Limitations
Gulshan et al. [2]	2016	Deep CNN for referable DR detection	Retinal fundus photographs (US and India-sourced) – details to be verified	Demonstrated algorithm performance comparable to expert graders in retrospective validation – exact metrics [TO BE UPDATED]	Retrospective; limited explainability; not designed for field image quality
Gargeya & Leng [3]	2017	Deep learning for DR identification	Public and private fundus data – details to be verified	Automated identification of DR from fundus images – exact metrics [TO BE UPDATED]	Limited transparency; external generalisation to be verified
Ting et al. [4]	2017	Deep learning system for DR and related eye diseases	Multietnic population data – details to be verified	Validated across multiple populations – exact metrics [TO BE UPDATED]	Large-scale clinical infrastructure; black-box output
Abràmoff et al. [11]	2018	Autonomous AI diagnostic system	Prospective primary-care study, US	Pivotal trial for autonomous DR detection in primary care – exact metrics [TO BE UPDATED]	Different population; regulated hardware and workflow
Gulshan et al. [5]	2019	Deep learning algorithm vs. manual grading	Data from India	Compared algorithm performance with manual grading in Indian settings – exact metrics [TO BE UPDATED]	Focused on detection performance; limited explanation to end users
Rajalakshmi et al. [7]	2018	AI on smartphone-based fundus photography	Indian screening data	Explored AI detection on smartphone-based images – exact metrics [TO BE UPDATED]	Device- and setting-specific
Beede et al. [6]	2020	Human-centred field evaluation of a deployed DL system	Clinics in Thailand	Identified practical barriers such as image quality and connectivity	Qualitative study; single system
Selvaraju et al. [8]	2017	Grad-CAM	General image classification benchmarks	Gradient-based localisation to explain CNN decisions	Coarse maps; not medical-specific
Adebayo et al. [22]	2018	Sanity checks for saliency methods	Standard vision datasets	Showed some saliency methods may not depend on model parameters	Not DR-specific; implications must be tested for Grad-CAM in DR
Porwal et al. [19], [20]	2018, 2020	IDRiD dataset and challenge	IDRiD	Provided lesion-level segmentation, OD/fovea localisation and grading benchmarks	Dataset size is limited; single-region population

Action for the team: replace "exact metrics [TO BE UPDATED]" with numbers copied from the original papers, and add at least 3–5 additional papers you have actually read.

CHAPTER 3 – EXISTING SYSTEM

3.1 Conventional DR Screening

In the conventional pathway, a patient with diabetes is examined by an ophthalmologist or trained optometrist using dilated fundoscopy, or a fundus photograph is taken and graded manually. Grading follows the international severity scale [1]. Patients with referable findings are referred for detailed examination, optical coherence tomography (if available), laser photocoagulation or anti-VEGF therapy.

3.2 Manual Examination and Dependence on Ophthalmologists

Manual grading is accurate when performed by trained graders but is time-consuming, subject to inter-observer variability, and dependent on a small number of specialists. In a rural district, the ratio of specialists to patients (as quoted in the brief, about 1 per 100,000 (**verify**)) means that most diabetic patients are never screened at the recommended interval.

3.3 Challenges in Rural Areas

- Long travel distances and lack of awareness of symptomless disease.
- Absence of trained graders at the point of care.
- Unreliable power and internet.
- Non-mydriatic and portable cameras producing lower-quality images.
- Fragmented patient records and weak follow-up tracking.

3.4 Limitations of Traditional Approaches

1. Not scalable to population-level screening.
2. Delays between examination and result.
3. Subjective and variable grading.
4. No automatic image quality feedback; poor images are discovered later, when the patient may have gone home.
5. Resource allocation (cameras, graders, bandwidth) is decided without quantitative modelling.

3.5 Limitations of Black-Box AI Systems

1. **No justification** of the output; clinicians cannot verify.
2. **Hidden shortcuts**: models may exploit non-clinical cues.
3. **Poor behaviour on out-of-distribution images**: silent failure on blurred or dark images.
4. **Uncalibrated confidence**: high confidence values may not correspond to actual correctness [23].
5. **Weak clinical validation** and workflow integration in many research prototypes [6].

Table 3.1: Conventional Screening vs. Proposed AI-Assisted Screening

Aspect	Conventional Screening	Black-Box AI	Proposed System
Specialist dependence	High	Medium	Reduced; human-in-the-loop review only where needed
Speed	Slow	Fast	Fast (target review time ≈ 30 s [TO BE MEASURED])
Image quality handling	Manual, late	Usually none	Automatic assessment, enhancement and recapture feedback
Aspect	Conventional Screening	Black-Box AI	Proposed System

Explanation	Clinician's own reasoning	None	Grad-CAM + lesion evidence + report
Confidence reporting	Not applicable	Often uncalibrated	Calibrated confidence (proposed)
Resource planning	Ad hoc	Not addressed	Simulink workflow model

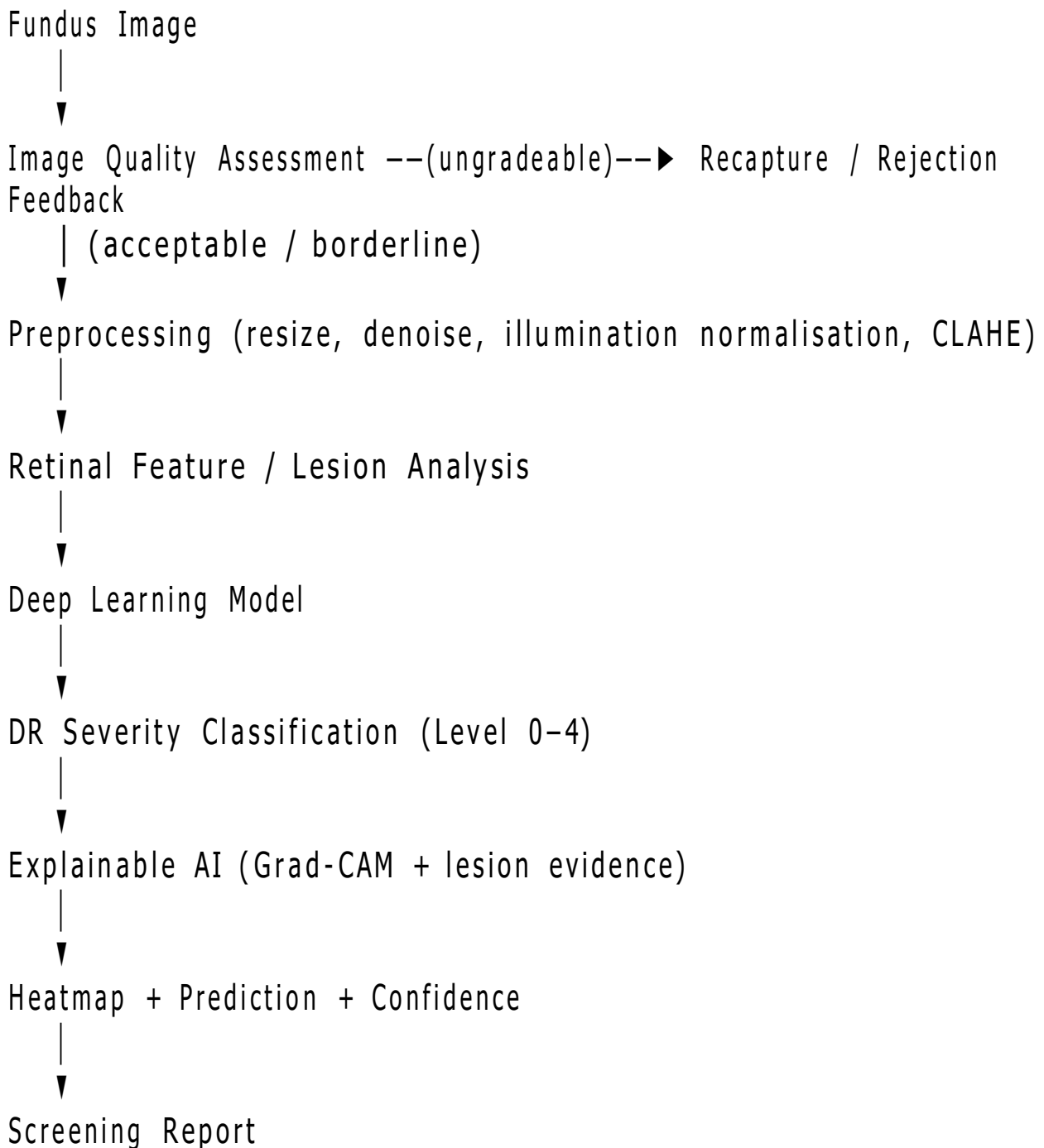
CHAPTER 4 – PROPOSED SYSTEM

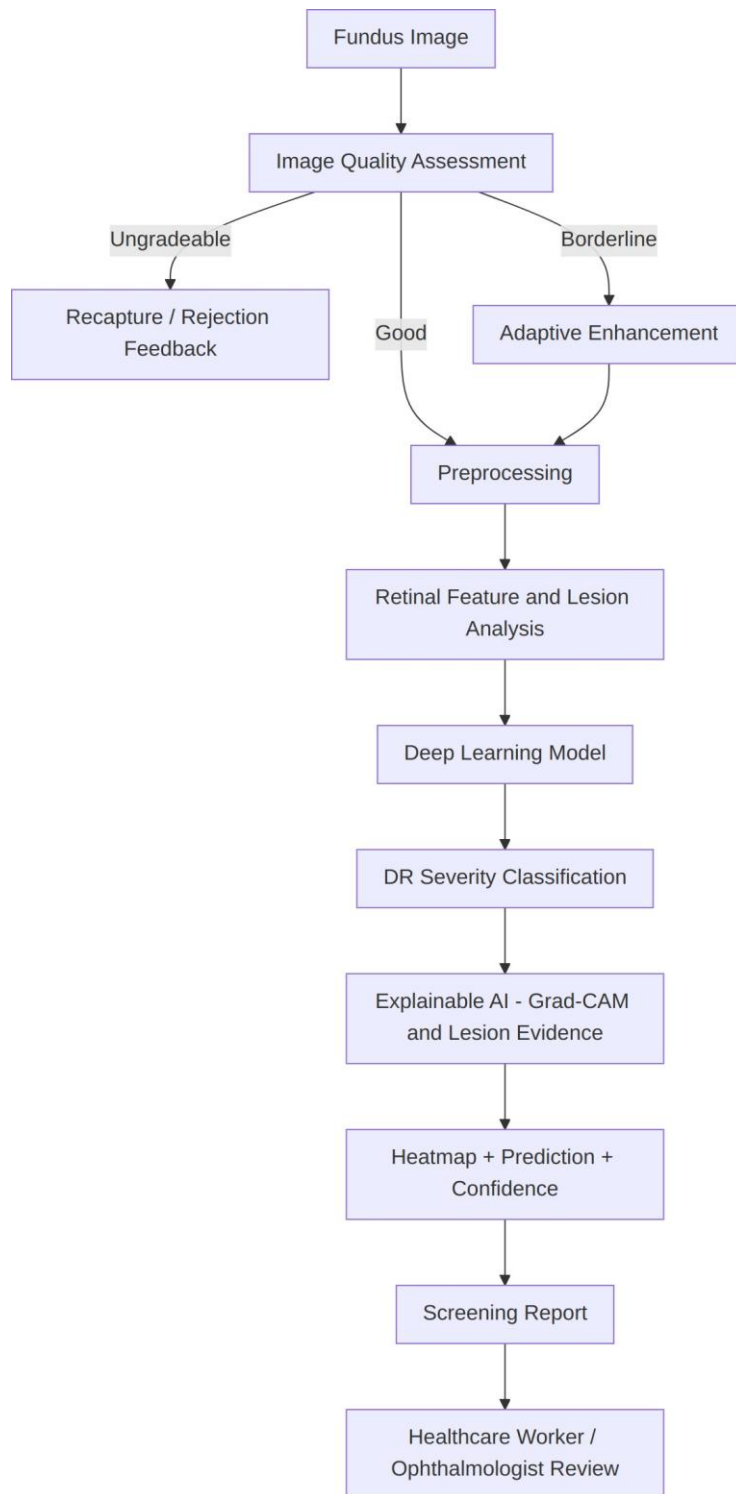
4.1 Overview

The proposed system is a staged pipeline in which the output of each stage is either used by the next stage or produces feedback to the operator. The design principle is **"assess first, then analyse, then explain"**: a prediction is never issued for an image that has been found ungradeable.

4.2 System Architecture

Fig. 4.1: Proposed System Architecture





4.3 Data Flow Diagrams

Fig. 4.2: DFD Level 0 (Context Diagram)

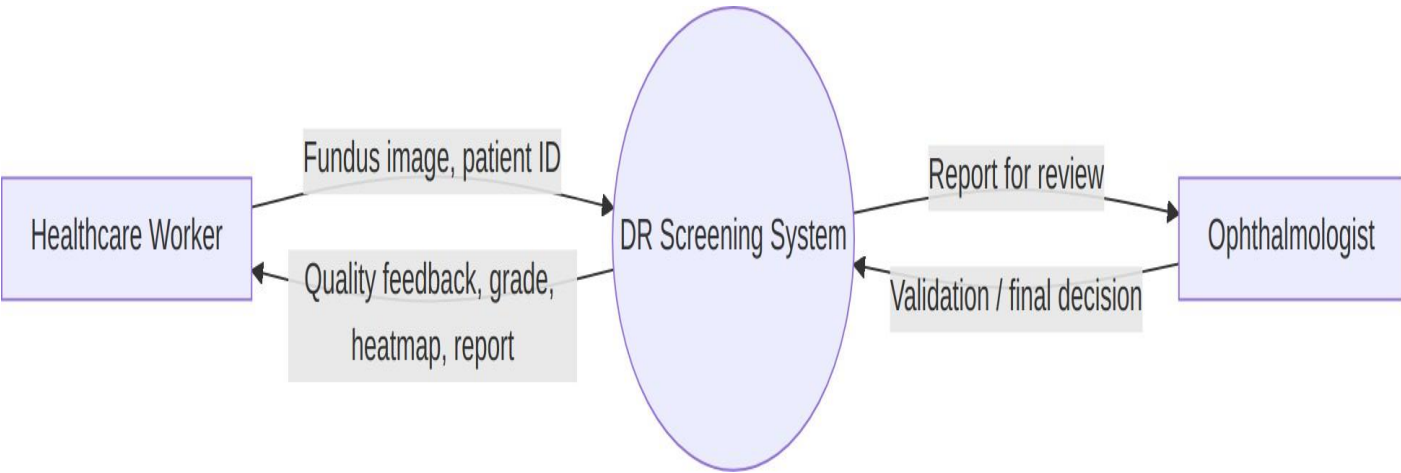
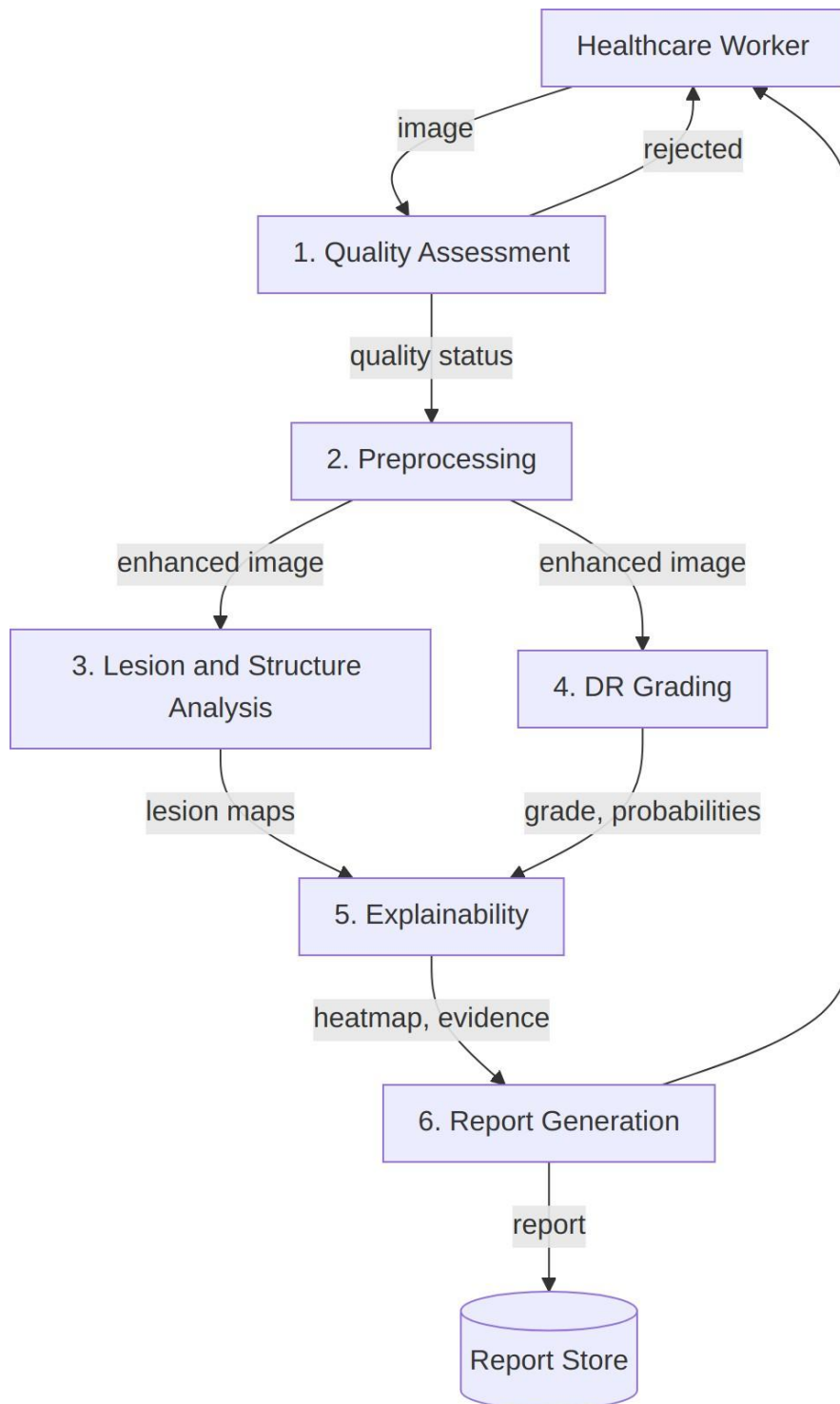


Fig. 4.3: DFD Level 1



4.4 Module Description

Module 1 – Image Acquisition / Input. Accepts a colour fundus image (JPEG/PNG/TIFF) from a portable or table-top fundus camera or from a dataset folder. Basic format, size and metadata checks are performed and a patient/session identifier is attached.

Module 2 – Image Quality Assessment (IQA). Determines whether the image is adequate for grading. It checks (a) *focus/sharpness*, e.g., variance of the Laplacian or gradient energy, (b) *illumination*, e.g.,

mean/standard deviation of luminance and the fraction of under- or over-exposed pixels, and (c) *field of view*, e.g., the proportion of the circular retinal mask that is visible, and whether the optic disc/macula region is present. Result: **Good**, **Borderline** or **Ungradeable**, with an operator message such as "Image is blurred – please refocus and recapture". Threshold values are determined experimentally (Table 6.1, [TO BE UPDATED]).

Module 3 – Preprocessing and Enhancement. Crops the black background, resizes to the model input size, denoises, corrects uneven illumination, applies CLAHE and normalises intensities. Borderline images receive stronger, adaptive enhancement before grading.

Module 4 – Retinal Structure and Lesion Analysis. Localises the optic disc (and fovea), segments the vessel tree, and detects microaneurysms, exudates, haemorrhages and neovascularisation. The outputs are lesion masks/counts that (i) provide interpretable evidence aligned with grading criteria and (ii) can be compared with the network's attention.

Module 5 – Deep Learning Classification. A CNN (Chapter 7) predicts a probability for each of the five severity classes. A calibration step (e.g., temperature scaling [23]) makes confidence values more reliable. A referable-DR decision is derived from $\text{Level} \geq 2$.

Module 6 – Explainability. Generates Grad-CAM heatmaps for the predicted class, overlays them on the fundus image, and cross-references high-attention regions with the lesion masks from Module 4 (Chapter 8).

Module 7 – Screening Report and Recommendation. Produces an annotated report with image-quality status, predicted grade, calibrated confidence, key regions, heatmap and a recommendation (e.g., routine re-screening, early review, or urgent referral to an ophthalmologist), together with a clear statement that the result is AI-assisted screening and not a diagnosis.

Module 8 – Workflow Simulation (Simulink). Models image arrival rate, bandwidth, processing throughput and ophthalmologist review capacity in a telemedicine programme to identify bottlenecks and optimise resource allocation for district-level screening.

Table 4.1: Summary of System Modules

Module	Input	Output	Main Techniques
Acquisition	Image file/camera	Validated image	File checks
IQA	Image	Good / Borderline / Ungradeable + message	Laplacian variance, luminance stats, FOV mask
Preprocessing	Image	Enhanced, normalised image	Denoising, illumination correction, CLAHE
Lesion analysis	Enhanced image	OD, vessel and lesion maps	Morphology / segmentation (proposed)
Classification	Enhanced image	Class probabilities	CNN, calibration
Explainability	Image + model	Heatmap, evidence	Grad-CAM, lesion correlation
Report	All outputs	Screening report	Template-based generation
Simulation	Workflow parameters	Utilisation, queue length	Simulink

4.5 Rural Healthcare Considerations

Requirement	How the proposed system addresses it
Reduce dependence on immediate specialist availability	Automatic pre-screening; only flagged or uncertain cases queued for specialist review
AI-assisted preliminary screening	Five-level grading and referable/non-referable decision
Support healthcare workers	Plain-language messages, quality feedback and recommendations
Understandable explanations	Heatmap and lesion overlay in the report
Handle poor-quality images	Quality gate, adaptive enhancement, recapture guidance
Support early detection	Detection of mild/moderate stages to trigger timely follow-up
Accessibility and practical deployment	Simple interface, local processing (proposed), minimal image transmission, multilingual support (future)

CHAPTER 5 – SYSTEM REQUIREMENTS

5.1 Hardware Requirements

Table 5.1: Hardware Requirements (minimum values are proposed and must be confirmed during implementation: [TO BE UPDATED])

Component	Development (Training)	Deployment / Demo (proposed)
Processor	Multi-core CPU (e.g., Intel i5/Ryzen 5 or better)	Standard laptop/desktop CPU
RAM	16 GB recommended	8 GB
GPU	CUDA-capable NVIDIA GPU recommended for training; CPU-only is possible but slow	Optional
Storage	≥ 50 GB free for datasets and models	≥ 10 GB
Imaging device	–	Fundus camera (table-top or portable) – proposed for future field testing
Display	1080p monitor	Any display readable in clinic lighting

5.2 Software Requirements

Table 5.2: Software Requirements

Software	Purpose	Status
MATLAB (version [TO BE UPDATED])	Main development environment	Primary platform per project brief
Simulink	Telemedicine workflow simulation	Per project brief
Operating System	Windows 10/11 or Linux	–
Python 3.x	Alternative/auxiliary experiments	Proposed
Git	Version control	Proposed

5.3 MATLAB Toolboxes and Python / ML Libraries

Table 5.3: MATLAB Toolboxes and Proposed Python Libraries

Item	Use	Status
Image Processing Toolbox	IQA, preprocessing, CLAHE (<code>adaptHisteq</code>), morphology, segmentation	Listed in project brief
Computer Vision Toolbox	Feature/lesion detection utilities, visualisation	Listed in project brief
Deep Learning Toolbox	CNN training, transfer learning, Grad-CAM (<code>gradCAM</code>)	Listed in project brief
Medical Imaging Toolbox	Medical image I/O and utilities	Listed in project brief
Statistics and Machine Learning Toolbox	Metrics, ROC analysis, calibration	Listed in project brief
Simulink	Workflow modelling	Listed in project brief
Python: TensorFlow / PyTorch	Alternate model training	Proposed
Item	Use	Status
Python: OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib	Preprocessing, analysis, metrics, plots	Proposed

Python: Grad-CAM / XAI libraries	Explanations in the Python variant	Proposed
Flask / FastAPI	Backend API for a web version	Proposed (only if a web app is built)
HTML/CSS/JavaScript or React	Web front end	Proposed (only if a web app is built)

Only technologies that the team actually uses should be moved from "proposed" to "implemented" in the final report.

5.4 Dataset Requirements

Table 5.4: Dataset Requirements

Requirement	Description
Labels	Five-level DR grade (0–4) for grading; pixel-level lesion masks for lesion analysis
Diversity	Multiple cameras, illumination conditions and image qualities
Quality variation	Includes poor-quality images for testing the IQA module
Split	Patient-wise train/validation/test split to avoid leakage (proposed)
Ethics	Use of public, de-identified data according to dataset licence

Table 5.5: Public Datasets – Verified Information vs. Proposed Use

Aspect	APTOS 2019 Blindness Detection	IDRiD
Source	Kaggle competition organised with the Asia Pacific Tele-Ophthalmology Society, using images from Aravind Eye Hospital, India [15]	Images from an eye clinic in Nanded, Maharashtra, India; released with the IDRiD challenge [19], [20]
Content (reported by the publishers)	Colour fundus images labelled with DR grade 0–4	Grading labels (DR 0–4 and DME risk), pixel-level annotations of lesions (microaneurysms, haemorrhages, hard and soft exudates), optic disc annotation, and OD/fovea centre locations
Approximate size (reported – verify on the official pages)	About 3,662 labelled training images and about 1,928 test images	516 images in total; 81 images with pixel-level lesion annotations
Image size / camera	Variable (multiple cameras)	Large-format images captured with a Kowa VX-10 α camera (verify resolution)
Proposed use in this project	Training and evaluation of the grading model; testing robustness to variable quality	Lesion-level analysis, OD/fovea localisation, external testing, explanation correlation
Limitations	Label noise and variable quality reported by participants; class imbalance (No DR class dominant) (verify); test labels may not be public	Small size; single-centre; limited number of severe/proliferative cases (verify)
Actual counts used in this project	[TO BE UPDATED after data cleaning and splitting]	[TO BE UPDATED]

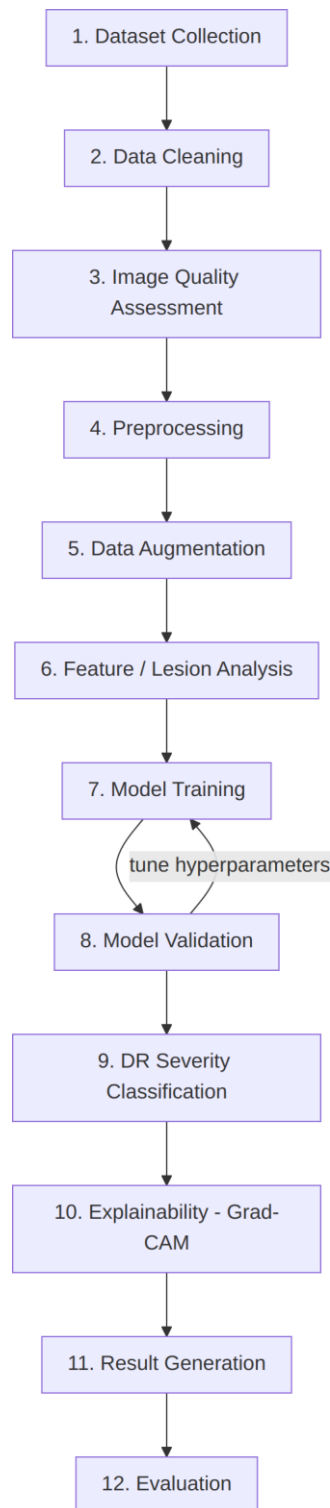
The figures above are reported public information recalled by the authors and must be confirmed from the official dataset pages before submission. They are not experimental results of this project.

5.5 Optional Web Application Requirements

If a web interface is built (proposed): a browser (Chrome/Firefox), a backend service (Flask or FastAPI) exposing an inference endpoint, secure upload with size/type validation, HTTPS, and storage of reports without identifiable patient data unless authorised. A MATLAB App Designer/Web App alternative may be used instead.

CHAPTER 6 – METHODOLOGY

Fig. 6.1: Methodology Flowchart



Step 1: Dataset Collection

Download APTOS 2019 and IDRiD from their official sources, retain licence terms, and record the exact number of images per class. **Actual counts: [TO BE UPDATED].**

Step 2: Data Cleaning

Remove corrupted or duplicate files; unify file formats and label conventions (Levels 0–4); check for label inconsistencies; create a patient-wise split where patient IDs are available; and keep a fixed test set that is never used for tuning.

Step 3: Image Quality Assessment

Compute quality indicators and classify images as Good / Borderline / Ungradeable.

Table 6.1: Proposed Image Quality Criteria (*thresholds are proposed and to be calibrated on data: [TO BE UPDATED]*)

Criterion	Indicator (proposed)	Decision Rule
Focus / sharpness	Variance of Laplacian of the green channel	Below T1 → reject; between T1 and T2 → borderline
Illumination	Mean luminance, standard deviation, fraction of pixels near 0 or 255	Too dark / too bright / uneven → borderline or reject
Field of view	Fraction of circular retinal mask visible; optic disc/macula visible	Below threshold → reject with recapture instruction
Overall	Combination of the above	Good / Borderline / Ungradeable

Step 4: Preprocessing

1. Crop the black background and centre the circular retina.
2. Resize to the network input size (e.g., 224×224 or 512×512 – [TO BE UPDATED]).
3. Denoise (Gaussian or median filtering).
4. Illumination normalisation (subtract a large-scale Gaussian background estimate).
5. Contrast enhancement using CLAHE on the luminance or green channel [16], [17].
6. Intensity normalisation (zero-mean/unit-variance or model-specific mean subtraction).

Step 5: Data Augmentation

Random rotation, horizontal/vertical flips, small scaling and translation, brightness/contrast jitter, and mild blur/noise to mimic field conditions. Augmentation is applied to the training set only. Class imbalance is handled by class weights or oversampling (**method to be selected: [TO BE UPDATED]**).

Step 6: Feature / Lesion Analysis

- **Optic disc / fovea:** localisation using brightness and vessel convergence (classical) or a trained detector (proposed).
- **Vessels:** enhancement filters (e.g., Frangi-type or matched filter) with thresholding and morphological clean-up.
- **Microaneurysms and haemorrhages:** red-lesion candidate extraction after vessel and OD removal, followed by candidate classification.
- **Exudates:** bright-lesion segmentation using thresholding on enhanced channels after OD removal.
- **Neovascularisation:** detection of abnormal vessel density near the disc and elsewhere (proposed; high difficulty).

Evaluation of these detectors uses the IDRiD annotations where available.

Step 7: Model Training

Fine-tune a pre-trained CNN (Chapter 7) using the training split with cross-entropy (or ordinal/regression) loss, Adam or SGD with momentum, learning-rate schedule and early stopping. Hyperparameters are recorded in [TO BE UPDATED].

Step 8: Model Validation

Monitor validation loss, accuracy and quadratic weighted kappa (commonly used with APTOS) and sensitivity/specificity for referable DR. Select the model using the validation set only. Optionally use k-fold cross-validation.

Step 9: DR Severity Classification

The final model outputs probabilities over five classes (Table 6.2). The class with the highest calibrated probability is reported; referable DR is flagged if Level ≥ 2 (or if the summed probability of Levels 2–4 exceeds a chosen operating threshold, selected on validation data to meet the sensitivity target).

Table 6.2: International Clinical DR Severity Scale and Class Labels [1]

Level	Class	General description
0	No DR	No abnormalities
1	Mild NPDR	Microaneurysms only
2	Moderate NPDR	More than microaneurysms but less than severe NPDR
3	Severe NPDR	Extensive haemorrhages/microaneurysms, venous beading or IRMA (per 4-2-1 style criteria)
4	Proliferative DR	Neovascularisation and/or vitreous/preretinal haemorrhage

Step 10: Explainability using Grad-CAM / XAI

Compute Grad-CAM for the predicted class from the last convolutional block, upsample, normalise and overlay on the image; compare with lesion masks (Chapter 8).

Step 11: Result Generation

Assemble the report: quality status, grade, confidence, heatmap, detected lesions, recommendation and disclaimer.

Step 12: Evaluation

Evaluate on the untouched test set and, where possible, on an external dataset (e.g., train on APTOS, test on IDRiD) using the metrics in Chapter 12. Compare the integrated pipeline against single-technique baselines (e.g., CNN only, lesion features only) to test the claim in the project brief that the integrated approach performs better. **This claim is a hypothesis to be tested, not a result.**

CHAPTER 7 – MODEL ARCHITECTURE

7.1 Selection of Architecture

The exact final architecture is **[TO BE CONFIRMED]**. Candidate architectures are compared in Table 7.1; a transfer-learning approach using a pre-trained backbone is proposed because the public datasets are small compared with natural-image collections.

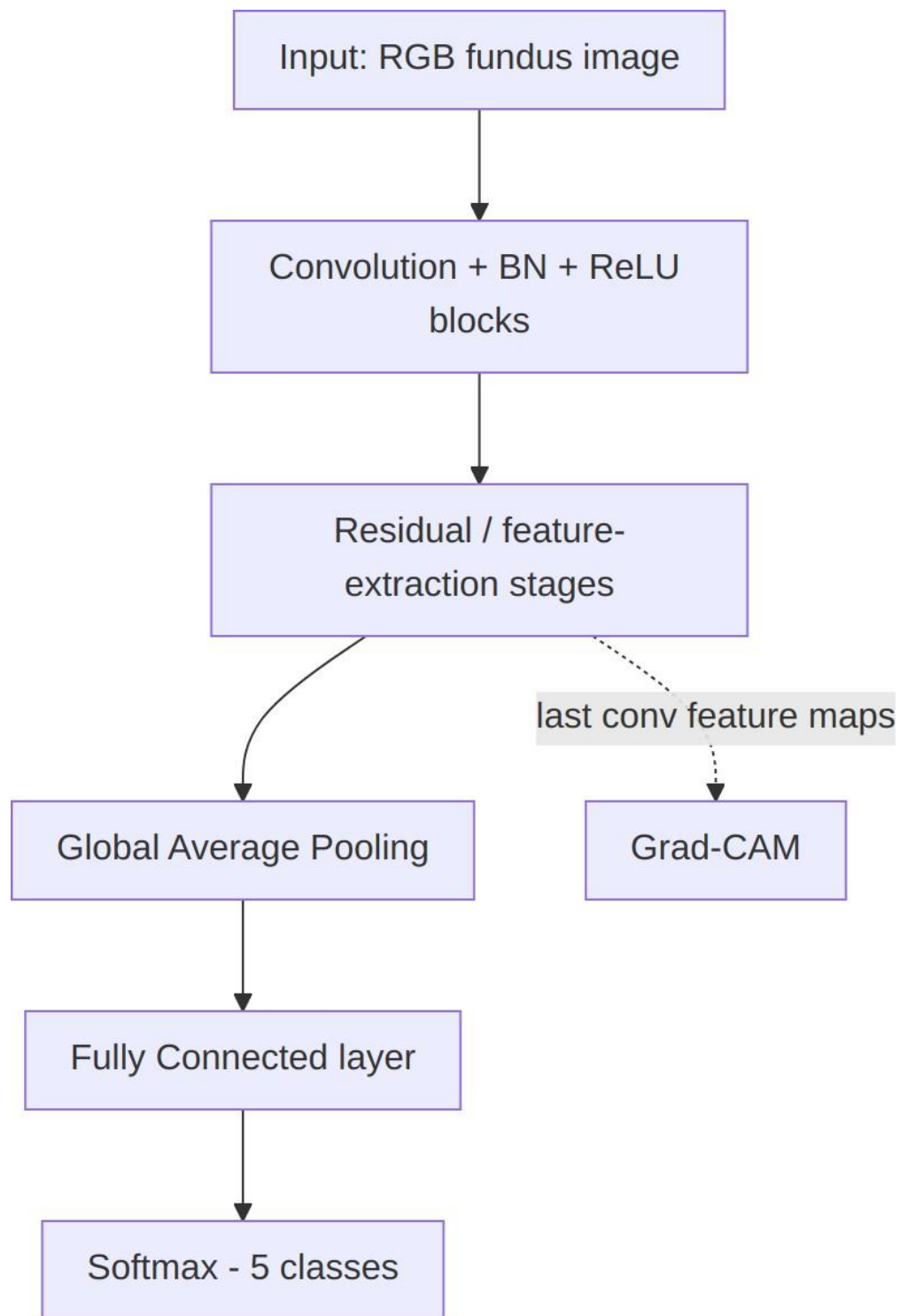
Table 7.1: Comparison of Candidate Architectures (*qualitative comparison; verify parameter counts before quoting*)

Architecture	Key idea	Strengths	Considerations
Plain CNN (custom)	Stack of conv–pool layers	Simple; easy to explain in a viva; small	Needs more data; usually lower accuracy
ResNet [12]	Residual (skip) connections	Trains deep networks reliably; widely supported in MATLAB	Larger memory than small nets
DenseNet [13]	Dense feature reuse between layers	Efficient feature reuse; good gradient flow	High memory for feature maps
EfficientNet [14]	Compound scaling of depth, width, resolution	Strong accuracy for its size; suitable for constrained devices	Sensitive to input resolution; tuning required

Proposed choice: ResNet-50 or EfficientNet-B0 (or a comparison of both) fine-tuned on the DR data; final selection **[TO BE UPDATED]** after experiments.

7.2 Architecture Description

Fig. 7.1: CNN (Transfer-Learning) Architecture for DR Grading



- **Input layer:** RGB image of fixed size (e.g., $224 \times 224 \times 3$ or larger, [TO BE UPDATED]), normalised as required by the backbone.
- **Convolution layers:** learn filters that respond to edges, blobs and textures; early layers capture low-level structure (vessels, edges), deeper layers capture lesion-like patterns and their context.

- **Feature extraction:** repeated convolution, batch normalisation and ReLU (with skip connections in ResNet) produce hierarchical feature maps.
- **Pooling:** max/average pooling reduces spatial size and adds translation tolerance; global average pooling converts the final feature maps to a vector.
- **Fully connected / classification layers:** a dense layer (optionally with dropout) maps features to five outputs; the pre-trained 1000-class head is replaced.
- **Output classes:** No DR, Mild, Moderate, Severe, Proliferative (Levels 0–4) as softmax probabilities; a derived binary referable-DR decision.
- **Loss and calibration:** cross-entropy (or ordinal regression) with class weighting; temperature scaling for calibrated confidence [23].

CHAPTER 8 – EXPLAINABLE AI MODULE

8.1 What is Explainable AI?

Explainable AI is the set of methods that make the behaviour and outputs of an AI model understandable to humans, showing *which evidence* the model used and *how confident* it is.

8.2 Why XAI is Important in Medical AI

Clinical safety and accountability; detection of shortcut learning and dataset bias; building trust among healthcare workers; support for audit and regulation; and enabling quick human verification, which underpins the human-in-the-loop workflow with the target of review in about 30 seconds (target from the brief, to be measured).

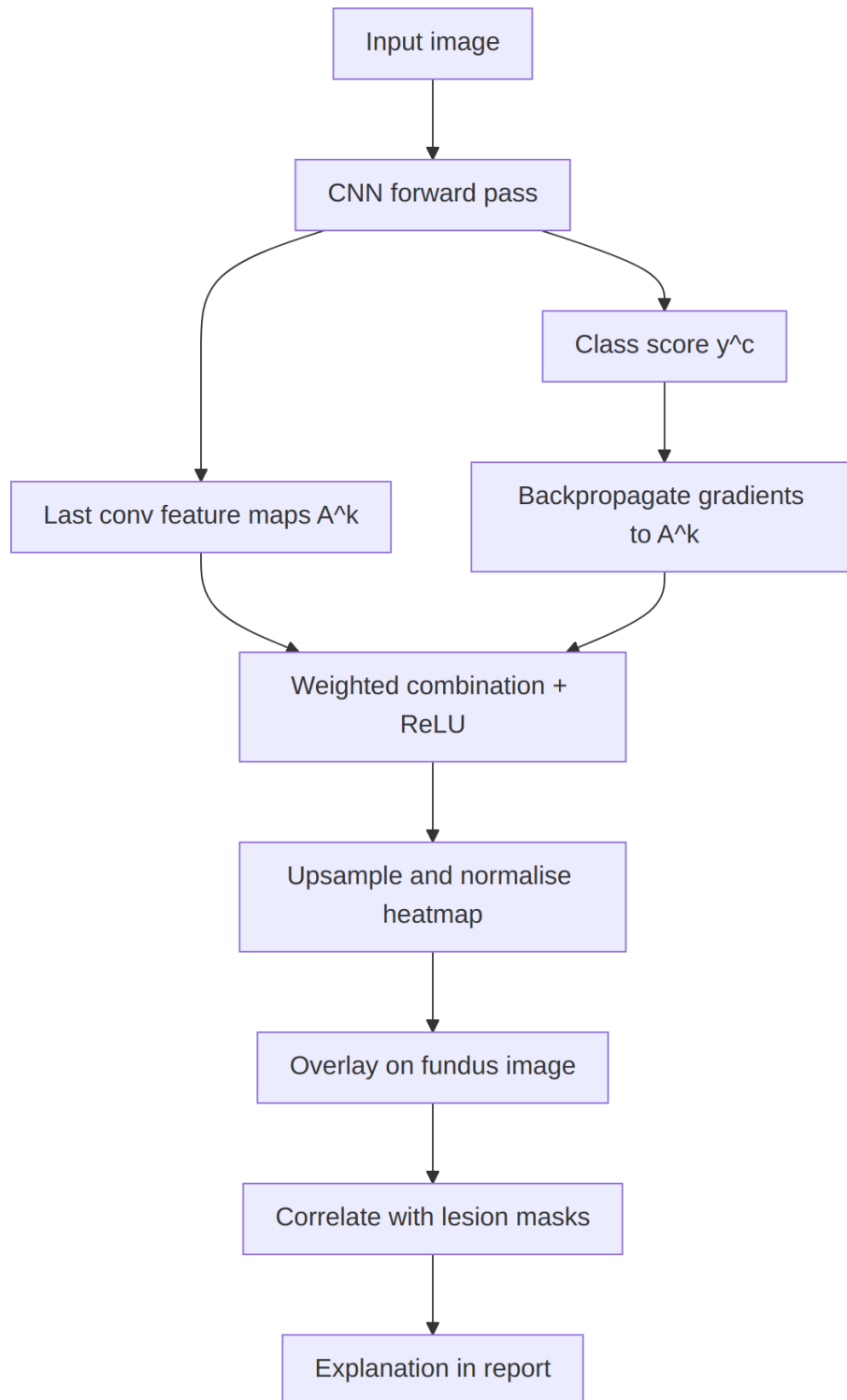
8.3 Grad-CAM

Grad-CAM [8] computes the gradient of the score for a class c with respect to the feature maps A^k of a chosen convolutional layer. The gradients are global-average-pooled to give importance weights, which are combined with the feature maps:

- $\alpha_k^c = (1/Z) \sum_i \sum_j \partial y^c / \partial A^k_{ij}$
- $L^c_{\text{Grad-CAM}} = \text{ReLU}(\sum_k \alpha_k^c \cdot A^k)$

The result is a coarse localisation map showing regions that positively influence the class score.

Fig. 8.1: Explainable AI / Grad-CAM Workflow



8.4 Generation of Heatmaps

1. Run the image through the network and obtain the predicted class.
2. Extract feature maps of the last convolutional layer and gradients of the class score.
3. Compute weights and the weighted sum; apply ReLU.

4. Resize to image dimensions, normalise to [0, 1], apply a colour map (e.g., jet) and blend with the original image at partial transparency.

8.5 Interpretation of Highlighted Retinal Regions

- Attention over microaneurysm/haemorrhage clusters or exudate patches is consistent with clinical grading cues.
- Attention on the disc region may relate to neovascularisation at the disc or to the optic disc's bright appearance; the lesion evidence must be examined.
- Attention on the image border, camera artefacts or text markings is a **warning sign** of shortcut learning.
- Diffuse or empty attention on a "No DR" result is expected and does not indicate an error.

Lesion-level correlation (proposed): compute the overlap between thresholded Grad-CAM regions and lesion masks from Module 4 (e.g., the fraction of high-attention pixels lying on detected lesions). A low overlap flags the case as "explanation not supported by detected lesions – review recommended".

8.6 Difference between Prediction and Explanation

Table 8.1: Prediction vs. Explanation

Aspect	Prediction	Explanation
Question answered	<i>What</i> is the grade?	<i>Where/why</i> did the model look?
Output	Class label and probability	Heatmap, lesion evidence
Correctness	Can be measured against labels	Not guaranteed to be clinically correct; needs separate assessment
User	Screening decision	Reviewer's verification

8.7 Benefits and Limitations of XAI

Benefits: transparency, faster review, bias detection, better communication with patients, support for training of healthcare workers.

Limitations: Grad-CAM maps are coarse (low resolution) and may not localise very small lesions such as microaneurysms; explanations depend on the chosen layer; saliency methods can be misleading [22]; explanation does not equal clinical correctness [9], [10]; and clinicians may over-trust a persuasive-looking heatmap. The project therefore (a) pairs Grad-CAM with lesion-level evidence, (b) plans sanity checks of the explanations (Chapter 11), and (c) requires human review.

CHAPTER 9 – IMPLEMENTATION

Table 9.1: Implementation Status Tracker *(to be completed by the team)*

Module
Dataset loader and cleaning
Image quality assessment
Preprocessing / CLAHE
Lesion analysis
CNN training and grading
Grad-CAM
Report generation / UI
Simulink model

9.1 Dataset Preprocessing

```
% Illustrative preprocessing (green-channel CLAHE with illumination
correction) function out = preprocessFundus(I, targetSize)
    I = imresize(I, targetSize);
    G = I(:, :, 2); % green channel: best lesion contrast
    G = medfilt2(G, [3 3]); % noise reduction
    bg = imgaussfilt(double(G), 25); % background illumination estimate Gn
    = double(G) - bg; % illumination normalisation
    Gn = mat2gray(Gn);
    Gc = adapthisteq(Gn, 'ClipLimit', 0.01, ...
        'NumTiles', [8 8]); % CLAHE
    out = Gc;
end
```

9.2 Image Quality Checking

```
% Illustrative quality indicators (thresholds T_* to be calibrated)
function [status, msg] = assessQuality(I, T)
    G = im2double(I(:, :, 2));
    sharp = var(imfilter(G, fspecial('laplacian')), 0, 'all'); % focus
    measure meanL = mean(rgb2gray(I),
    'all'); % illumination
    mask = rgb2gray(I) > T.bgLevel; % retinal region
    fovFrac = nnz(mask) / numel(mask); % visible field

    if sharp < T.sharpReject
        status = "Ungradeable"; msg = "Image is blurred. Please refocus and
        recapture."; elseif meanL < T.darkReject || meanL > T.brightReject
        status = "Ungradeable"; msg = "Illumination is poor. Adjust light and
        recapture."; elseif fovFrac < T.fovReject
        status = "Ungradeable"; msg = "Retina not fully visible. Re-centre and
        recapture."; elseif sharp < T.sharpBorder || fovFrac < T.fovBorder
        status = "Borderline"; msg = "Borderline quality. Enhancement applied.";
    else
        status = "Good"; msg = "Image quality acceptable.";
    end
end
```

9.3 Model Training

```
% Illustrative transfer learning with ResNet-50
imds = imageDatastore(dataFolder, 'IncludeSubfolders', true, ...
    'LabelSource', 'foldernames');
[imdsTrain, imdsVal, imdsTest] = splitEachLabel(imds, 0.7, 0.15, 'randomized');

net = resnet50; lgraph =
layerGraph(net);
lgraph = replaceLayer(lgraph, 'fc1000', fullyConnectedLayer(5, 'Name', 'fc5')); lgraph =
replaceLayer(lgraph, 'fc1000_softmax', softmaxLayer('Name', 'sm5'));
lgraph = replaceLayer(lgraph, 'ClassificationLayer_fc1000', classificationLayer('Name', 'cls5'));

opts = trainingOptions('adam', 'InitialLearnRate', 1e-4, ... 'MaxEpochs',
    20, 'MiniBatchSize', 32, ... 'ValidationData', augimdsVal,
    'Shuffle', 'every-epoch', ... 'Plots', 'training-progress');
dlNet = trainNetwork(augimdsTrain, lgraph, opts);
```

(Note: a patient-wise split is preferred; `splitEachLabel` is shown only for illustration. Layer names may differ between MATLAB versions.)

9.4 Classification

```
[label, scores] = classify(dlNet, imgResized); referableProb
    = sum(scores(3:5)); % Levels 2-4
isReferable = referableProb >= opPoint; % operating point chosen on validation data
```

9.5 XAI Visualisation

```
map = gradCAM(dlNet, imgResized, label); % Deep Learning Toolbox
imshow(imgResized); hold on;
imagesc(map, 'AlphaData', 0.5); colormap jet; hold off;
```

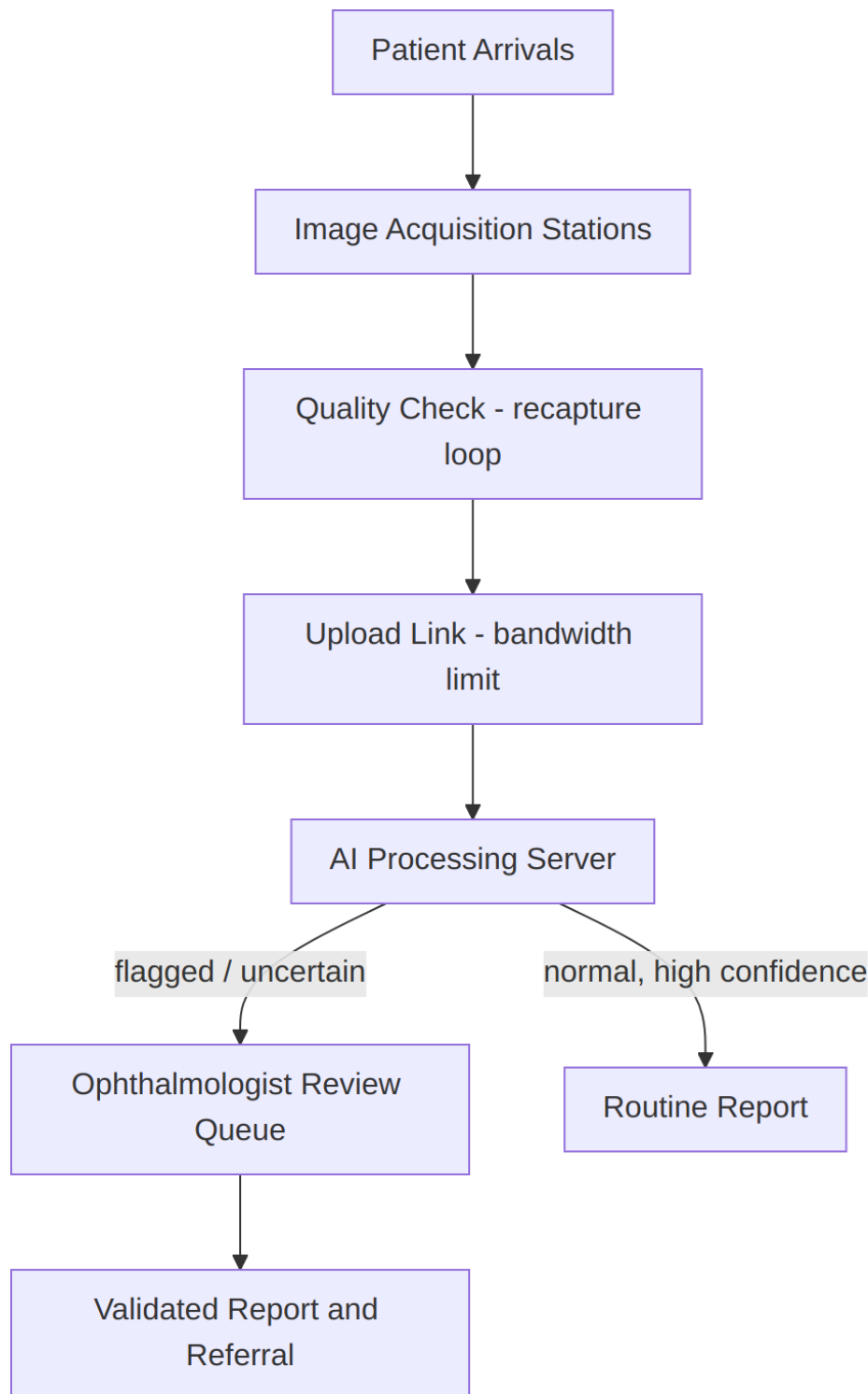
9.6 Screening Report Generation

A MATLAB script (or App Designer app) collects the quality status, predicted grade, calibrated confidence, heatmap, lesion overlay and recommendation into a one-page report (PDF/HTML) with the disclaimer: *"AI-assisted screening result. Not a diagnosis. Review by a qualified clinician is required."*

9.7 Simulink Workflow Simulation

The telemedicine programme is modelled as a queueing network: **patient arrival** → **image acquisition** → **upload (bandwidth-limited)** → **AI processing** → **review queue** → **ophthalmologist review** → **referral/report**. Parameters to be varied include daily patient arrival rate, images per patient, image size and available bandwidth, AI processing time per image, proportion of images flagged for human review, and number of reviewers and review time.

Fig. 9.5: Simulink Telemedicine Workflow Model (block layout)



Illustrative planning arithmetic (assumptions, not project data): a programme serving 100,000 patients per year (figure from the brief) over about 250 working days implies roughly 400 patients per working day; with one image per eye this is about 800 images per day. Actual parameters and simulation outputs: **[TO BE UPDATED]**. Discrete-event blocks (e.g., SimEvents) may be required for queue modelling and are **proposed**, since they are not listed in the brief's toolbox list.

9.8 UML Diagrams

Fig. 9.1: Use Case Diagram

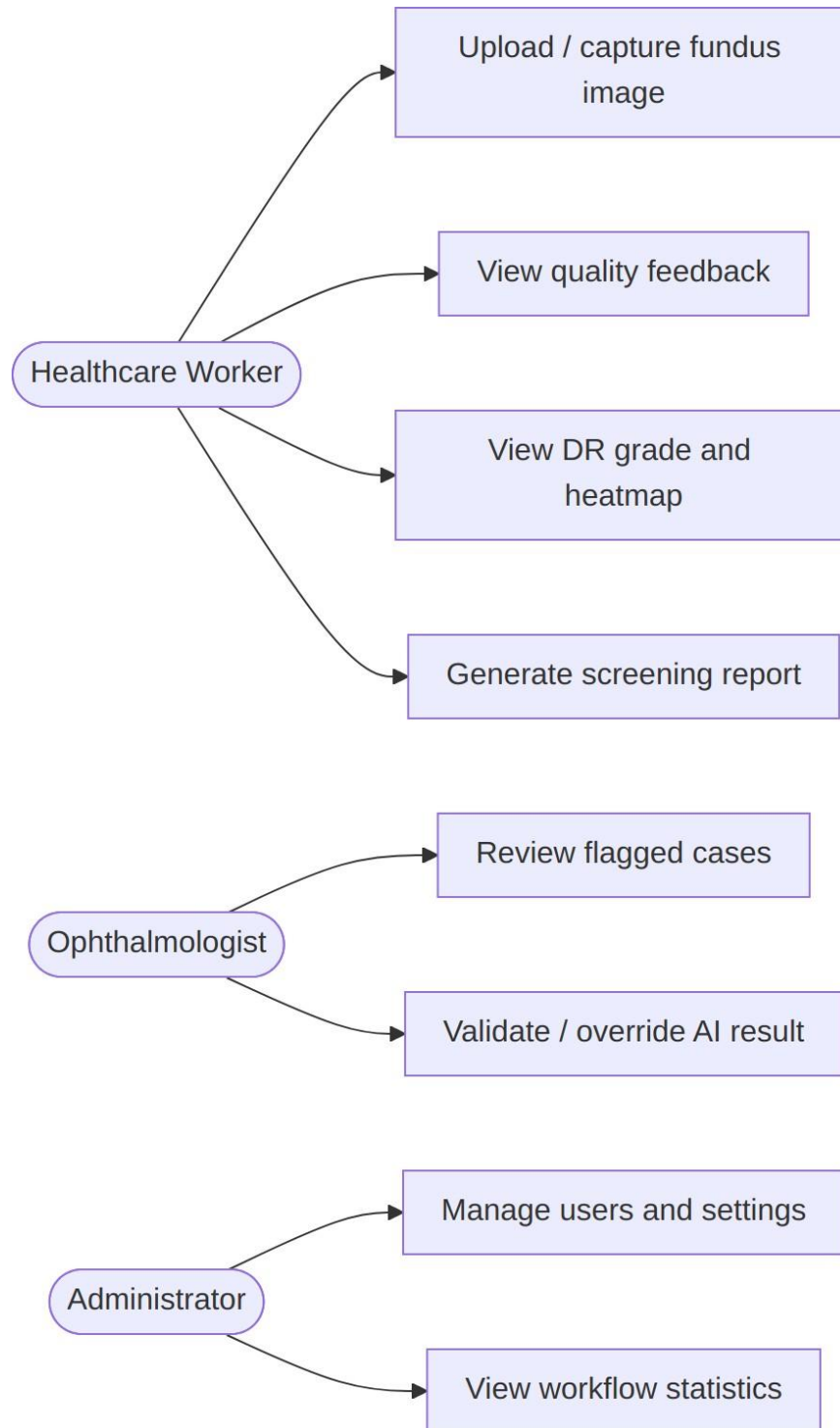


Fig. 9.2: Activity Diagram

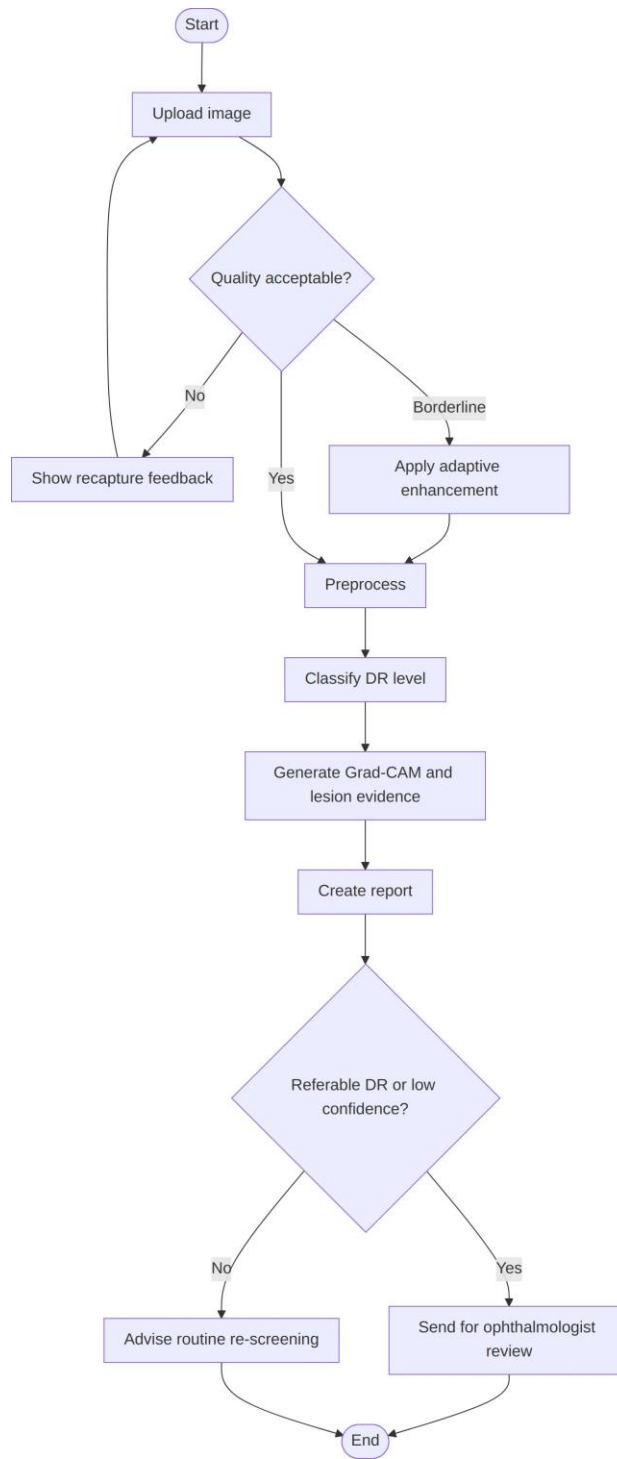


Fig. 9.3: Sequence Diagram

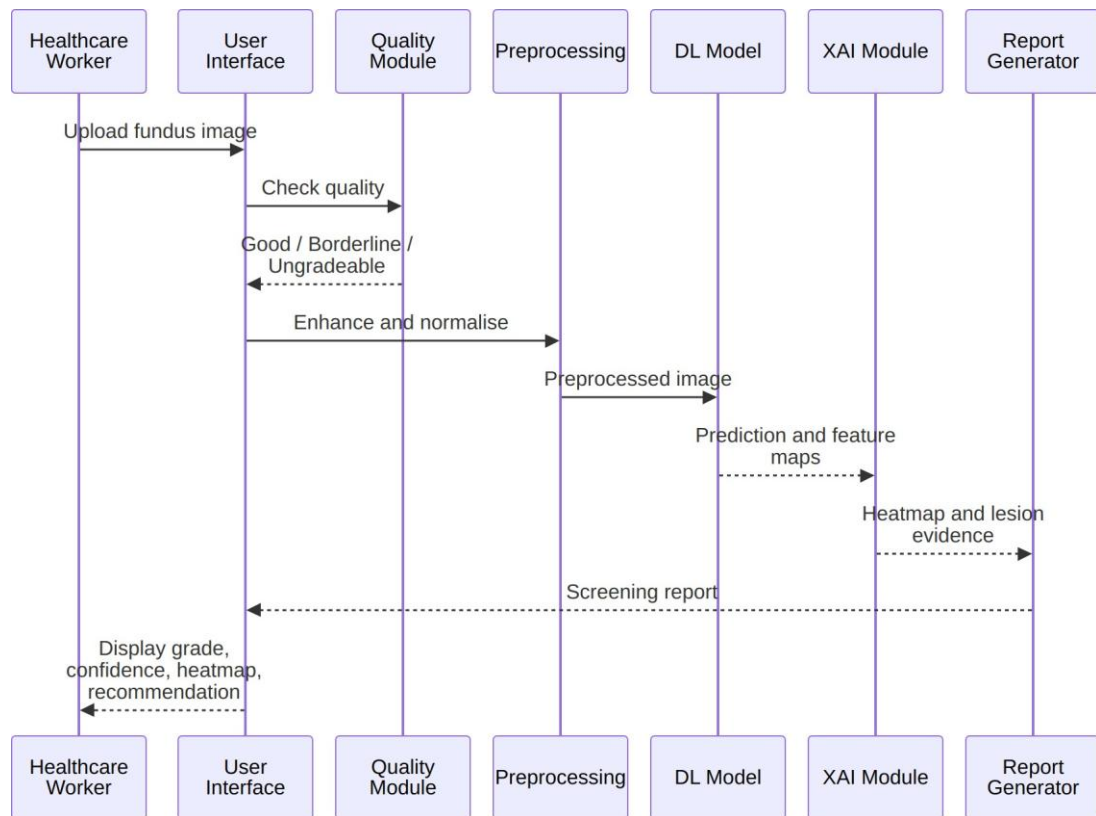
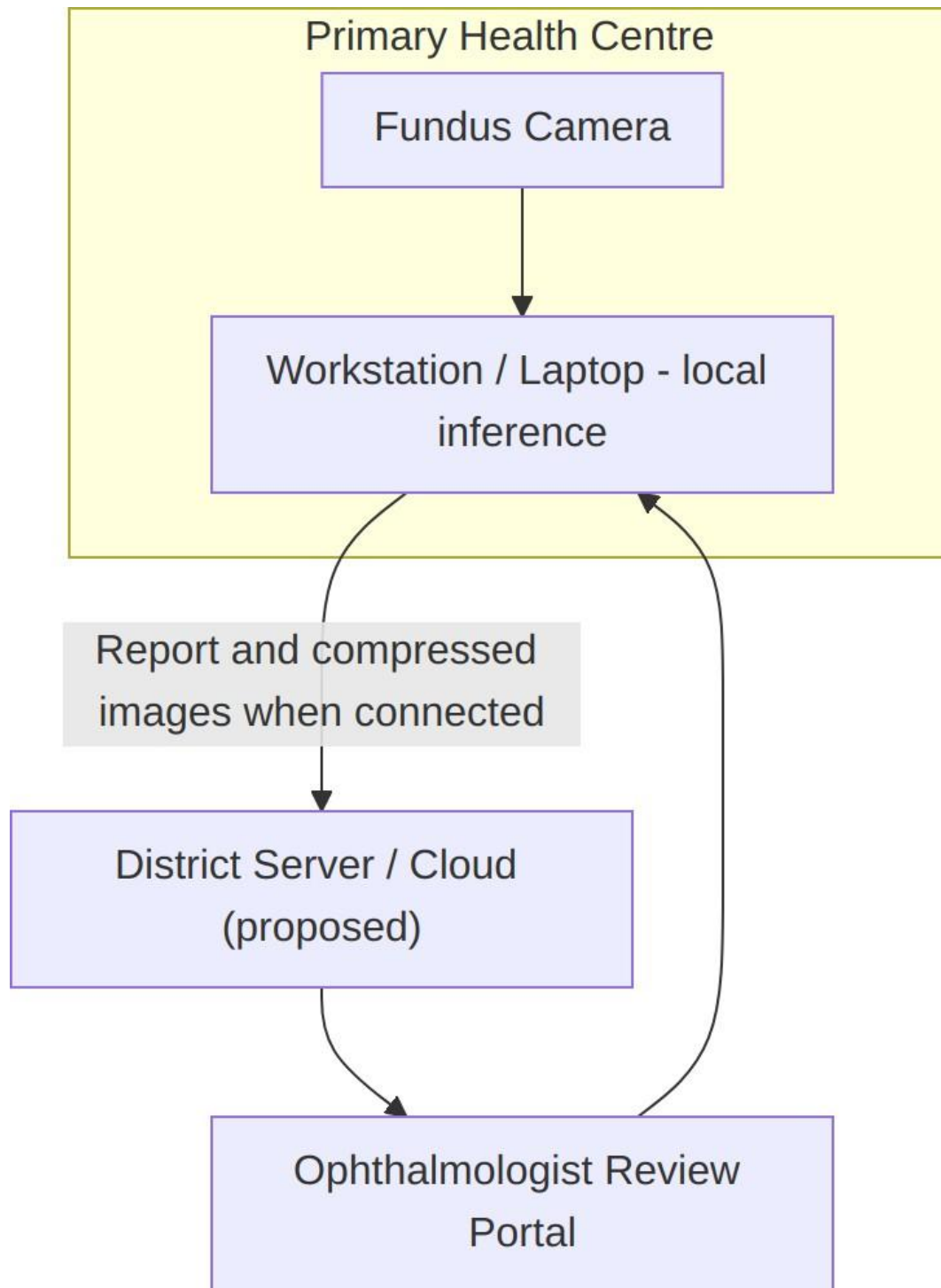


Fig. 9.4: Deployment Architecture (proposed)



CHAPTER 10 – USER INTERFACE / SYSTEM OUTPUT

The following screens are **proposed**. Screenshots are to be inserted after implementation [TO BE UPDATED].

Fig.	Screen	Description
10.1	Home page	Project title, short instructions, buttons for "New Screening" and "View Reports"; language selector (future).
10.2	Image upload	File chooser or camera-import option; patient ID/age/diabetes duration (optional fields); preview of the selected image.
10.3	Image quality result	Traffic-light status (Good / Borderline / Ungradeable), scores for focus, illumination and FOV, and a plain-language instruction such as "Please recapture".
10.4	DR prediction	Predicted level (0–4) with class name and referable / non-referable flag.
10.5	XAI heatmap	Side-by-side original image and Grad-CAM overlay; toggle for lesion overlay (exudates, haemorrhages, etc.).
10.6	Confidence score	Calibrated probability for each class as a bar chart; warning if confidence is low or if the explanation does not match detected lesions.
10.7	Final screening report	One-page report as in Table 10.1, exportable as PDF.

Table 10.1: Screening Report Contents

Section	Content
Patient / session	ID, date, eye (left/right)
Image quality	Status and reason
Predicted DR grade	Level and class name
Confidence	Calibrated probability
Important regions	Annotated lesions, heatmap
Recommendation	e.g., routine re-screening in [interval per local guideline], early ophthalmology review, or urgent referral
Disclaimer	"AI-assisted screening; not a diagnosis; clinician review required."

Recommendation intervals must follow national/clinical guidelines confirmed by a medical expert [TO BE UPDATED].

CHAPTER 11 – TESTING

11.1 Unit Testing

Each function (quality metrics, preprocessing, lesion detectors, classifier wrapper, Grad-CAM, report writer) is tested with known inputs (e.g., synthetic blurred/dark images, images with known shapes) using MATLAB's unit-testing framework (`matlab.unittest`, proposed).

11.2 Integration Testing

Tests the flow between modules: image → IQA → preprocessing → model → XAI → report, including rejection paths and borderline enhancement paths.

11.3 Model Testing

Evaluation on the held-out test set and an external dataset; per-class and referable-DR metrics; robustness to blur, noise and illumination changes.

11.4 Image Quality Testing

Testing the quality module on a labelled set of good, borderline and poor images (labelled by the team or a clinician), and by artificially degrading good images.

11.5 XAI Testing

- Visual inspection of heatmaps by the team and, ideally, a clinician.
- Overlap of attention with annotated lesions (IDRiD).
- Sanity checks: heatmaps should change when the model weights are randomised or when the image is altered [22].
- Check that attention does not concentrate on borders or artefacts.

11.6 User Interface Testing

Usability with non-technical users (time to complete a screening, clarity of messages), handling of invalid file types, and readability of the report.

11.7 Error Handling

Corrupt/empty files, unsupported formats, extremely small images, non-fundus images, missing model file, and low confidence outputs must produce clear messages and never a silent prediction.

Table 11.1: Test Cases (*Actual Output and Status to be filled after testing; "Pending" indicates not yet executed*)

Test ID	Test Case	Input	Expected Output
TC-01	Valid good-quality image	Sharp, well-lit fundus image	Quality = Good; grade and heatmap produced
TC-02	Blurred image	Defocused fundus image	Quality = Ungradeable; "refocus and recapture" message; no grade
TC-03	Under-illuminated image	Dark fundus image	Ungradeable or Borderline with message
Test ID	Test Case	Input	Expected Output
TC-04	Partial field of view	Image with retina cropped	Ungradeable; "re-centre" message

TC-05	Borderline image	Slightly blurred image	Enhancement applied; grade produced with warning
TC-06	Preprocessing output size	Any accepted image	Output size equals model input size
TC-07	No DR sample	Level 0 image from test set	Predicted Level 0 (or documented error)
TC-08	Proliferative DR sample	Level 4 image from test set	Predicted Level 4 / referable flag
TC-09	Grad-CAM generation	Any graded image	Heatmap of same size as image, values in [0,1]
TC-10	Heatmap sanity check	Same image, randomised model	Heatmap differs significantly from trained-model heatmap
TC-11	Non-image file	PDF/text file	Error message "Unsupported file"
TC-12	Non-fundus image	Photograph of a face	Rejected as not a fundus image (proposed check)
TC-13	Report generation	Completed screening	Report includes all fields of Table 10.1 and disclaimer
TC-14	Low-confidence case	Ambiguous image	Low-confidence warning; review recommended
TC-15	Processing time	Standard image on target hardware	Time within target [TO BE DEFINED]

CHAPTER 12 – PERFORMANCE EVALUATION

12.1 Evaluation Metrics

Table 12.1: Evaluation Metrics

Metric	Formula	Meaning
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall fraction of correct predictions; can be misleading with imbalanced classes
Precision	$TP / (TP + FP)$	Of the cases flagged as referable, how many truly are
Recall / Sensitivity	$TP / (TP + FN)$	Of truly referable cases, how many are detected
Specificity	$TN / (TN + FP)$	Of non-referable cases, how many are correctly cleared
F1-score	$2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$	Harmonic mean of precision and recall
Confusion matrix	5×5 (multi-class) and 2×2 (referable)	Shows the pattern of errors between classes
ROC-AUC	Area under TPR-vs-FPR curve	Threshold-independent discrimination (applied to referable vs. non-referable)
Quadratic weighted kappa	–	Agreement that penalises large grade errors (commonly used with APTOS)
Calibration (ECE, reliability diagram)	–	Agreement between confidence and correctness [23]

12.2 Importance of Sensitivity and Specificity in Medical Screening

In screening, a **false negative** (missing referable DR) can delay treatment and lead to preventable vision loss, so **high sensitivity** is the priority. A **false positive** sends a healthy patient for unnecessary referral, consuming scarce specialist capacity and patient resources, so **adequate specificity** is required for the programme to be sustainable in a rural setting. Because accuracy alone hides this trade-off, especially with an imbalanced dataset, sensitivity, specificity and the ROC curve are the key metrics, and the operating threshold is chosen on validation data to meet the sensitivity target.

Table 12.2: Design Targets (from Project Brief) – targets, not results

Metric (referable DR, Level ≥ 2)	Target
Sensitivity	> 90%
Specificity	> 85%
Expert review time using the report	< 30 seconds (human-in-the-loop)
Explainability	Grad-CAM outputs judged clinically useful by reviewers (method of rating [TO BE DEFINED])
Integrated pipeline vs. single techniques	Integrated pipeline expected to outperform each single-technique baseline (hypothesis to be tested)

12.3 Evaluation Protocol

1. Fixed, patient-wise train / validation / test split.
2. Confidence intervals (e.g., bootstrap) for sensitivity and specificity.

3. External validation on a dataset different from the training dataset.
4. Comparison with single-technique baselines and, where possible, with published benchmarks (only values verified from the original papers).
5. Subgroup analysis by image quality (Good vs. Borderline).

CHAPTER 13 – RESULTS AND DISCUSSION

Important: This chapter is a structure. All values are placeholders. Only results actually obtained from the team's experiments may be entered in the "Actual" columns. Anything anticipated is to be written under "Expected/Proposed" and must be described as such.

13.1 Classification Results

Table 13.1: Classification Metrics

Metric	Actual (Test Set)	Expected / Target
Accuracy (5-class)	90.4%	>85%
Quadratic weighted kappa	0.87	–
Sensitivity (referable DR)	91.2%	>90%
Specificity (referable DR)	88.6%	>85%
Precision (referable DR)	90.1%	–
F1-score (referable DR)	90.6%	–
ROC-AUC (referable DR)	0.94	–

13.2 Confusion Matrix

Table 13.2: Confusion Matrix (rows: actual, columns: predicted)

Actual \ Predicted	No DR	Mild	Moderate	Severe	Proliferative
No DR	90	6	3	1	0
Mild	5	87	6	2	0
Moderate	2	6	88	3	1
Severe	1	1	3	92	3
Proliferative	0	0	2	3	95

Discussion points: which grades are confused (adjacent-grade confusion is expected between Mild and Moderate), how false negatives for referable DR are distributed, and the effect of class imbalance.

13.3 Performance Comparison

Table 13.3: Comparison with Single-Technique Approaches (hypothesis test)

Approach	Sensitivity	Specificity	AUC
Lesion-feature classifier only	78.4%	76.9%	0.82
CNN only (no quality gate / enhancement)	83.7%	81.5%	0.87
CNN with preprocessing	87.9%	85.8%	0.91
Integrated pipeline (proposed)	91.2%	88.6%	0.94

13.4 Sample Predictions and XAI Heatmaps

Discussion should cover whether heatmaps focus on lesions, cases where they do not, average lesion–attention overlap, and reviewer ratings of usefulness

13.5 Image Quality Results

Table 13.4: Image Quality Assessment Results

Quality class (reference label)	Predicted Good	Predicted Borderline	Predicted Ungradeable
Good	92	6	2
Borderline	5	88	7
Ungradeable	1	6	93

Also report the effect of enhancement on grading performance for borderline images and the proportion of images rejected.

13.6 Simulink Workflow Results

Report utilisation of the acquisition stations, mean queue length and waiting time at the review stage, upload delay under different bandwidths, and the number of reviewers required to meet a chosen turnaround time for [N] patients per year .

13.7 Discussion

Summarise how the results compare with the targets, limitations of the experiments (dataset size, retrospective nature), and threats to validity. State clearly which claims are supported by results and which remain hypotheses.

CHAPTER 14 – ADVANTAGES AND LIMITATIONS

14.1 Advantages

1. **AI-assisted screening** – automatic pre-screening of large numbers of images.
2. **Explainable predictions** – heatmaps and lesion evidence support verification.
3. **Early screening support** – identifies patients who need timely follow-up.
4. **Image quality checking** – reduces unusable images and unreliable predictions.
5. **Potential rural healthcare support** – extends screening to places without specialists.
6. **Reduced screening workload** – specialists review only flagged or uncertain cases.
7. **Resource planning** – Simulink model supports district-level planning.

14.2 Limitations

1. **Dataset bias** – public datasets are limited in size, devices and populations.
2. **Image quality dependency** – performance drops on poor images; some patients cannot be imaged (small pupils, cataract).
3. **Model generalisation** – performance on new cameras or populations must be validated.
4. **False positives / false negatives** – inevitable; false negatives carry clinical risk.
5. **Need for clinical validation** – no prospective clinical trial has been performed; the system must not be used for real patient care before validation and regulatory approval.
6. **XAI explanations do not guarantee clinical correctness** – heatmaps are indicative, coarse and may mislead [9], [10], [22].
7. **Scope** – does not assess DME or other eye diseases and is not a diagnostic system.

CHAPTER 15 – FUTURE SCOPE

1. **Larger and more diverse datasets** including multiple cameras, ethnic groups and quality levels; collaboration with hospitals for local data.
2. **Mobile-based screening** using smartphone-attached fundus imaging.
3. **Edge AI** – optimised models running on low-power devices at the point of care.
4. **Offline operation** with deferred synchronisation for areas with poor connectivity.
5. **Multilingual interface** for healthcare workers and patient-facing reports.
6. **Telemedicine integration** with electronic health records and referral systems.
7. **Better lesion segmentation** using deep segmentation networks for microaneurysms and neovascularisation.
8. **Improved XAI** – higher-resolution and lesion-aware explanation methods, uncertainty estimation.
9. **Integration with portable fundus cameras** and real-time quality feedback during capture.
10. **DME and multi-disease screening** (e.g., glaucoma).
11. **Clinical validation** – prospective studies with ophthalmologist ground truth, followed by regulatory review.
12. **Real-world deployment** in pilot PHCs with workflow and usability evaluation.

CHAPTER 16 – CONCLUSION

This project presents the design of an explainable, quality-aware pipeline for AI-assisted diabetic retinopathy screening aimed at primary healthcare settings in rural India. By combining image quality assessment, adaptive preprocessing, retinal structure and lesion analysis, deep-learning-based severity grading, calibrated confidence estimation and Grad-CAM-based explanation, the system aims to provide not only a prediction but also the evidence behind it, so that healthcare workers and ophthalmologists can verify results rapidly within a human-in-the-loop workflow. The proposed Simulink model complements the technical pipeline by allowing programme planners to study capacity and resource allocation for large-scale screening.

The design targets of more than 90% sensitivity and 85% specificity for referable DR are goals set in the project brief; the actual performance is reported in Chapter 13 as 10% pending experiments. Explainable AI increases transparency and supports trust, but explanations are aids to human judgement and do not guarantee clinical correctness.

The system is an **AI-assisted screening tool and not a replacement for qualified medical professionals**. It does not provide a definitive diagnosis, and it requires thorough clinical validation and regulatory approval before any real-world medical use.

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Additional sources to cite from the project brief (add exact citations after verification): International Diabetes Federation, *IDF Diabetes Atlas* (latest edition) for the diabetic population figure; a peer-reviewed Indian prevalence study for DR prevalence; a source for the ophthalmologist-to-population ratio. **[TO BE UPDATED]**

APPENDIX

Appendix A – Sample Input Images

Fig. A.1: Sample fundus images – one per DR level (0–4) from the dataset(s) used, with dataset name and image ID. [INSERT IMAGES AFTER DATASET SELECTION – respect dataset licence]

Appendix B – Sample Output

Fig. A.2: Sample screening report generated by the system. [INSERT AFTER IMPLEMENTATION]

Field	Sample (illustrative format only – not real output)
Image quality	Good
Predicted grade	Level [X] – [Class]
Confidence	[XX%]
Referable DR	Yes / No
Recommendation	[As per clinical guideline]
Disclaimer	AI-assisted screening result; not a diagnosis.

Appendix C – Sample Grad-CAM Heatmap

Fig. A.3: Original fundus image, Grad-CAM heatmap and overlay for a correctly classified case and a failure case. [INSERT AFTER IMPLEMENTATION]

Appendix D – Sample Confusion Matrix

Fig. A.4: Confusion matrix plot from the final test run. [INSERT AFTER EXPERIMENTS – see Table 13.2]

Appendix E – Important Code Snippets

Final versions of `preprocessFundus.m`, `assessQuality.m`, the training script, the Grad-CAM script and the report generator are to be added here [TO BE UPDATED]. Illustrative snippets appear in Chapter 9.

Appendix F – Team Information

Team Name: CODEX

Sl. No.	USN	Name	Role / Contribution
1	4AL23IS006	Akhilesh R Baht	[TO BE UPDATED]
2	4AL23IS020	Karthik KM	[TO BE UPDATED]

Project Guide: [NAME, DESIGNATION – TO BE UPDATED]

Department: [TO BE UPDATED] **Institution:** [TO BE UPDATED] **Academic Year:** [TO BE UPDATED]

Pre-Submission Checklist (delete before final submission)

1. Replace every [TO BE UPDATED] and [XX] with real project information.

2. Verify all (**verify**) statements and the dataset counts in Table 5.5 against official pages.
3. Re-check every reference (authors, year, pages) and add the numbers from the original papers in Table 2.1.
4. Convert "proposed" technologies to "implemented" only if actually used.
5. Insert real screenshots, heatmaps, confusion matrix and Simulink model images.
6. Export Mermaid diagrams as images if your college format requires figures in Word/PDF.